VITALITY 16

Phentolamine

Alpha-adrenergic blocker that widens blood vessels; given by injection or as eye drops



CLASS Alpha blocker

STUDIED FOR Dental numbness; ED (off-label)

HUMAN EVIDENCE FDA approved (other uses); small ED trials

US STATUS FDA approved, not for ED

STUDIED FOR

- **Dental numbness** — HUMAN
- **Pheochromocytoma** — HUMAN
- **Erections (oral)** — HUMAN
- **Erections (injected)** — HUMAN

ED use is off-label. No FDA-approved phentolamine lists ED. Oral Vasomax and compounded Bimix and Trimix are not FDA approved.

HOW IT WORKS IN THE BODY

Phentolamine ▶ Alpha receptors ▶ Vessels widen ▶ Penile blood flow

- **Blocks alpha-adrenergic receptors,** so blood vessels widen
- **Penile arteries** appear mainly under alpha-adrenergic control
- Alpha blockade helped erections in **animal and human lab models**

EVIDENCE & WHAT'S KNOWN

EVIDENCE

FDA APPROVED
LEVEL 5 OF 5

- Approved: pheochromocytoma 1952, OraVerse 2008
- Oral ED Phase 3 trials known mainly via reviews
- Injected ED data: small studies of mixes

NOT YET ESTABLISHED

- Full oral Phase 3 trial reports

QUICK PRACTICAL INFO

HALF-LIFE
19 minutes in blood after a vein injection (label)

ROUTES STUDIED
Injection, dental injection, eye drops; by mouth in ED trials

PHARMACOKINETICS
About 13% of a vein dose in urine unchanged

SOLD AS
Prescription injection, OraVerse, Ryzumvi; Bimix and Trimix compounded

MONITORED IN TRIALS
Blood pressure, heart rate and rhythm

SPORT (WADA)
Not on the 2026 list

REGULATORY

FDA	Approved: injection 1952, OraVerse 2008, eye 2023
FDA	Not approved for ED; Bimix, Trimix compounded
WADA	Not on the 2026 list

REFERENCES

1. Hikma, phentolamine mesylate for injection label (DailyMed), Mar 2024
2. Goldstein et al., World J Urol, oral phentolamine review, Feb 2001
3. Bechara et al., J Urol, papaverine-phentolamine vs alprostadil, Jun 1997

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